

A Project Report On
“Smart Environmental Monitoring System Using IOT And Data Analytics”

Submitted in partial fulfillment of the requirements for the Award of Degree of

**BACHELOR OF TECHNOLOGY
ELECTRONICS & TELECOMMUNICATION ENGINEERING**

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CERTIFICATE

This is to certify that the project report entitled “**Smart Environmental Monitoring System using IOT and Data Analytics**” submitted by **Ms. Triveni Dhavandkar, Ms. Ashlesha Thakare, Ms. Pranita Bire, Ms. Monali Bhelkar, Ms. Anushka Deshpande** is work done by team and submitted in 6th semester during 2025-2026 academic year, in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY in ELECTRONICS & TELECOMMUNICATION ENGINEERING.**

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Amit Tripathi
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Project Name: “Smart Environmental Monitoring System using IOT and Data Analytics”

INTRODUCTION

Environmental pollution has become a critical global issue due to rapid industrialization, urban expansion, and increased vehicular emissions. The continuous release of harmful gases, particulate and heat into the environment has significantly affected air quality and overall ecological balance. Traditional environmental monitoring methods rely on manual data collection, which is time-consuming, prone to human error, and incapable of providing real-time insights.

To overcome these limitations, there is a growing need for an automated and intelligent monitoring system that can continuously track environmental conditions with high accuracy and efficiency.

This project presents a **Smart Environmental Monitoring System using Internet of Things (IoT) and Data Analytics**, designed to monitor key environmental parameters such as temperature, humidity, air quality, and light intensity in real time. The system uses multiple sensors integrated with a microcontroller to collect environmental data. This data is then processed and transmitted to a cloud platform via a wireless communication module, enabling remote access and continuous monitoring.

The IoT-based approach ensures that environmental data can be accessed from at any time through web or mobile applications. Additionally, the integration of basic data analytics allows the system to identify abnormal conditions and trigger alerts when predefined thresholds are exceeded. This makes the system highly useful in applications in smart cities, industrial safety, agriculture, and pollution control.

The proposed system is cost-effective, scalable, and easy to deploy, making it suitable for both small-scale and large-scale environmental monitoring applications.

Literature Survey

[1] INTERNATIONAL JOURNAL OF ADVANCED RESEARCH IN SCIENCE, COMMUNICATION AND TECHNOLOGY (IJARSCT)

This paper presents an IoT-based environmental monitoring system that uses multiple sensors to measure parameters such as temperature, humidity, and air quality. The system integrates sensors with a microcontroller to collect real-time data, which is then transmitted to a cloud platform using wireless communication. The proposed model emphasizes low cost, real-time monitoring, and ease of implementation. However, the system mainly focuses on basic monitoring and lacks advanced data analytics and intelligent alert mechanisms, which limits its effectiveness in critical applications.

[2] INTERNATIONAL JOURNAL OF ENGINEERING TECHNOLOGY AND MANAGEMENT SCIENCES

This research work discusses the development of a smart monitoring system using IoT and cloud platforms. The system collects environmental data through sensors and uploads it to a cloud server for visualization and analysis. The use of IoT enables remote monitoring and reduces manual effort. The design is simple and cost-effective, making it suitable for small-scale applications. However, the system is highly dependent on internet connectivity and does not provide accurate prediction or analysis of environmental changes.

[3] INTERNATIONAL JOURNAL OF PROGRESSIVE RESEARCH IN ENGINEERING MANAGEMENT AND SCIENCE (IJPREMS)

This paper proposes a wireless IoT-based system for monitoring environmental parameters using sensors and WiFi modules. The collected data is transmitted to a web interface where users can monitor conditions remotely. The system supports real-time updates and improves accessibility through mobile devices. Despite its advantages, the system lacks integration of multiple sensors for comprehensive monitoring and does not include advanced features such as automated decision-making or predictive analytics.

Objectives

The objective of developing a Smart Environmental Monitoring System using IoT and Data Analytics can be outlined as follows:

Real-Time Environmental Monitoring: Continuously monitor environmental parameters such as temperature, humidity, air quality, and light intensity using sensors to provide accurate and up-to-date data.

Remote Accessibility: Enable users to access environmental data remotely through cloud platforms or mobile applications without being physically present at the monitoring location.

Detection of Harmful Gases: Identify and measure the presence of hazardous gases in the environment using gas sensors, ensuring safety in industrial and urban areas.

Automated Alert System: Generate alerts when environmental parameters exceed predefined threshold values, allowing timely action to prevent hazardous situations.

Data Logging and Visualization: Store collected data on a cloud platform and present it in graphical format for easy analysis and understanding of environmental trends.

Cost-Effective Solution: Develop a low-cost system using easily available components such as Arduino and basic sensors, making it affordable for wide-scale implementation.

Ease of Implementation and Use: Design the system in a simple and user-friendly manner so that it can be easily installed, operated, and maintained without requiring advanced technical skills.

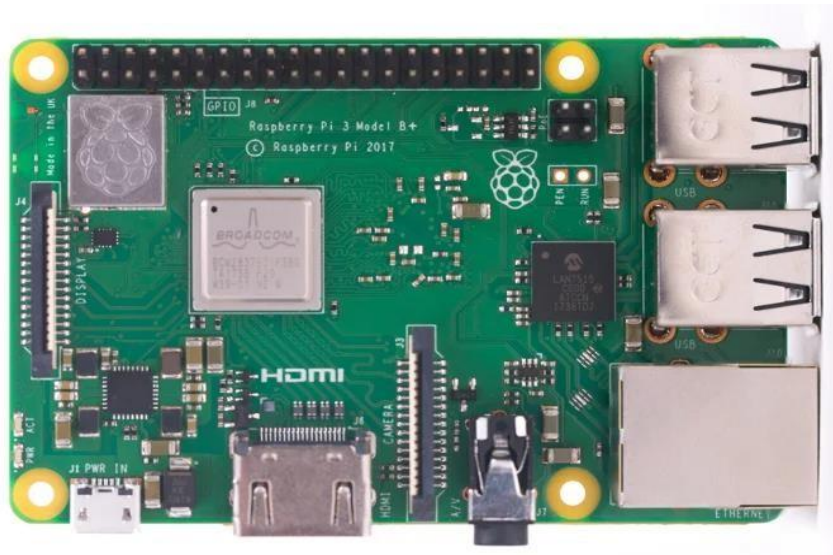
Integration with IoT Ecosystem: Enable communication between sensors, microcontrollers, and cloud platforms to create a connected system that supports automation and remote monitoring.

Scalability: Design the system in such a way that additional sensors and features can be integrated in the future without major modifications.

Energy Efficiency: Optimize power consumption to ensure stable and long-term operation, especially for continuous environmental monitoring applications.

Components used

- **Raspberry Pi:-**



(Fig.1)

- **ADC Module:-**



(Fig.2)

- **MQ135 Sensor:-**



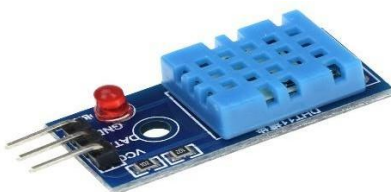
(Fig.3)

- **HDMI Cable:-**



(Fig.4)

- **DHT11 Sensor:-**



(Fig.5)

SOFTWARE:

:

```
import time

import board

import busio

import math

import adafruit_dht

import paho.mqtt.client as mqtt

# ThingSpeak MQTT Config

THINGSPEAK_MQTT_HOST = "mqtt3.thingspeak.com"

THINGSPEAK_CLIENT_ID = "NSogDx4KGhQIDAUqCxgUERA"

THINGSPEAK_USERNAME = "NSogDx4KGhQIDAUqCxgUERA"

THINGSPEAK_PASSWORD = "mFTIEHvmKdSBGQhBJTqbytFk"

CHANNEL_ID = "3318971"

# DHT22 Setup

dht_device = adafruit_dht.DHT22(board.D4)

# ADS1115 + MQ135 Setup

import adafruit_ads1x15.ads1115 as ads

from adafruit_ads1x15.analog_in import AnalogIn

i2c = busio.I2C(board.SCL, board.SDA)

adc = ads.ADS1115(i2c)

chan = AnalogIn(adc, 0)

# MQ135 Constants
```

```

RL = 10.0

VC = 5.0

CLEAN_AIR_FACTOR = 3.6

ALPHA = 0.02

#MQTT Setup

def on_connect(client, userdata, flags, rc):

    if rc == 0:

        print(" Connected to ThingSpeak")

    else:

        print("Connection failed:", rc)

client = mqtt.Client(client_id=THINGSPEAK_CLIENT_ID)

client.username_pw_set(THINGSPEAK_USERNAME,
THINGSPEAK_PASSWORD)

client.on_connect = on_connect

print("Connecting to ThingSpeak...")

client.connect(THINGSPEAK_MQTT_HOST, 1883, 60)

client.loop_start()

# Functions

def calculate_rs(voltage):

    return RL * ((VC / voltage) - 1)

def calculate_ppm(rs, r0):

    return 116.6020682 * ((rs / r0) ** -2.769034857)def get_aqi(ppm):

    if ppm <= 800:

```

```

        return 50, "Good"

    elif ppm <= 1200:

        return 100, "Satisfactory"

    elif ppm <= 2000:

        return 200, "Moderate"

    elif ppm <= 5000:

        return 300, "Poor"

    elif ppm <= 10000:

        return 400, "Very Poor"

    else:

        return 500, "Severe"

def average_rs(samples=10):

    values = []

    for _ in range(samples):

        v = chan.voltage

        if v > 0:

            values.append(calculate_rs(v))

        time.sleep(0.1)

    return sum(values) / len(values)

# Initial Calibration

print(" Calibrating MQ135...")

```

```

rs_avg = average_rs(30)

R0 = rs_avg / CLEAN_AIR_FACTOR

print(f'R0: {R0:.2f}')

# MAIN LOOP

while True:

    try:

        # ----- MQ135 -----

        rs = average_rs(5)

        ppm = calculate_ppm(rs, R0)

        # Auto-calibration

        R0 = (1 - ALPHA) * R0 + ALPHA * (rs / CLEAN_AIR_FACTOR)

        aqi, status = get_aqi(ppm)

    # DHT22

    try:

        temp = dht_device.temperature

        hum = dht_device.humidity

    except RuntimeError:

        temp, hum = None, None

    # PRINT

    print(f'Temp: {temp}°C | Humidity: {hum}%')

    print(f'PPM: {ppm:.2f} | AQI: {aqi} ({status})')

```

```

print("-----")

# MQTT SEND

if temp is not None and hum is not None:

    topic = f'channels/{CHANNEL_ID}/publish'

    payload = (

        f'field1={temp:.2f} &"

        f'field2={hum:.2f} &"

        f'field3={ppm:.2f} &"

        f'field4={aqi}"

    )

    client.publish(topic, payload)

    print( " Data sent to ThingSpeak")


# ThingSpeak free limit

time.sleep(20)


except KeyboardInterrupt:

    print("Stopped by user")

    break

```

```
except Exception as e:
```

```
    print("Error:", e)
```

```
    time.sleep(5)
```

```
# Cleanup
```

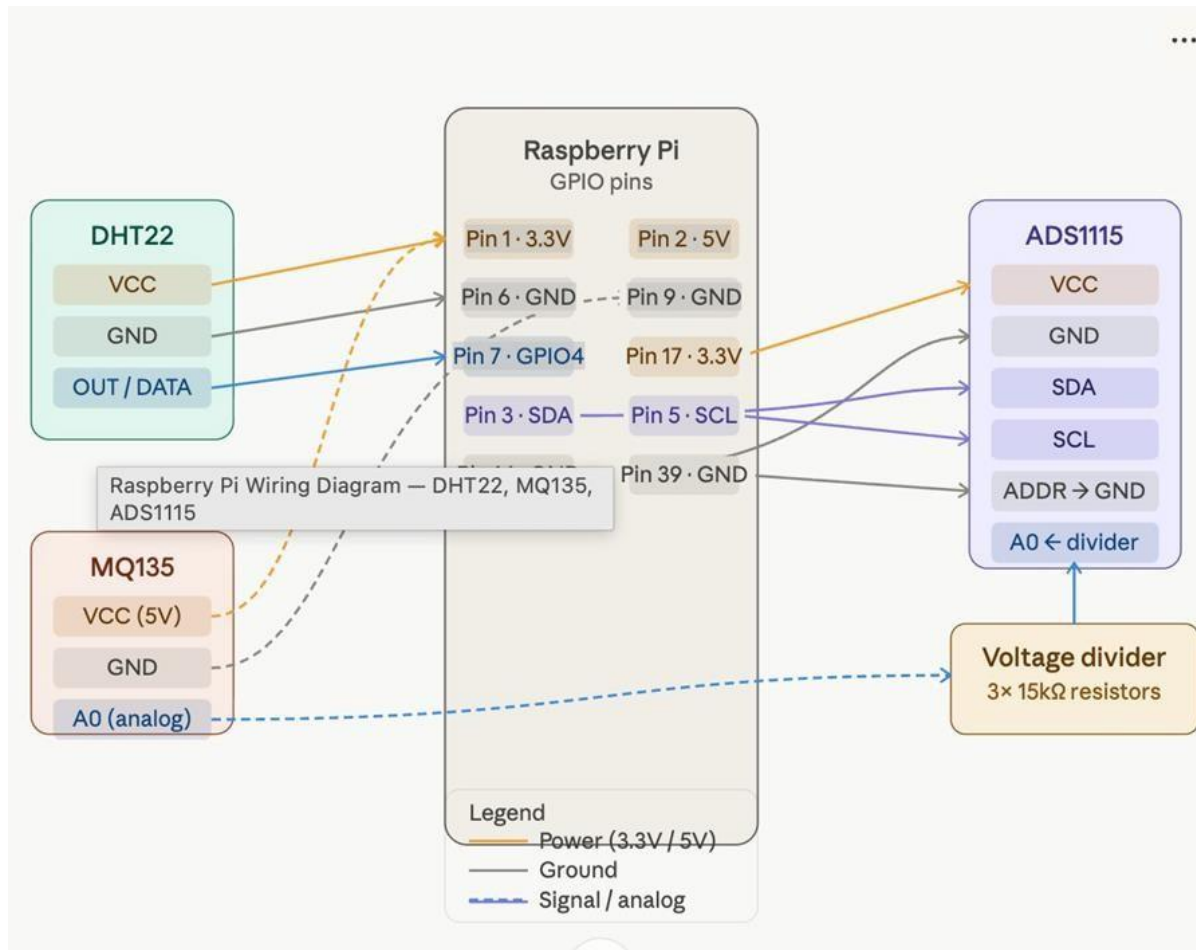
```
dht_device.exit()
```

```
client.loop_stop()
```

```
client.disconnect()
```

```
python thingspeak_all.py
```

Circuit Diagram



Working

The working of the Smart Environmental Monitoring System begins with powering the microcontroller and connected sensors. Once the system is initialized, all sensors such as the temperature and humidity sensor (DHT11), gas sensor (MQ series), and light sensor (LDR) start sensing environmental parameters from the surroundings.

The Arduino microcontroller continuously reads data from these sensors. The analog signals from the gas sensor and LDR are converted into digital values using the built-in ADC of the microcontroller, while the DHT11 directly provides digital output for temperature and humidity. These values are processed and formatted into meaningful units such as degree Celsius, percentage humidity, and gas concentration levels.

After processing, the collected data is transmitted to the cloud platform using a WiFi module such as ESP8266. The system establishes a connection with a WiFi network, and once connected, it sends the sensor data to the IoT cloud server at regular intervals. This enables real-time monitoring of environmental conditions from remote locations.

On the cloud platform, the data is displayed in graphical and numerical formats, allowing users to easily observe trends and variations in environmental parameters. The system also compares the sensor values with predefined threshold limits. If any parameter exceeds the safe range, an alert is generated to notify the user about the abnormal condition.

The entire process runs continuously in a loop, ensuring uninterrupted monitoring of the environment. In case of network failure or sensor malfunction, the system may temporarily stop transmitting data, and normal operation resumes once the connection is restored.

Output

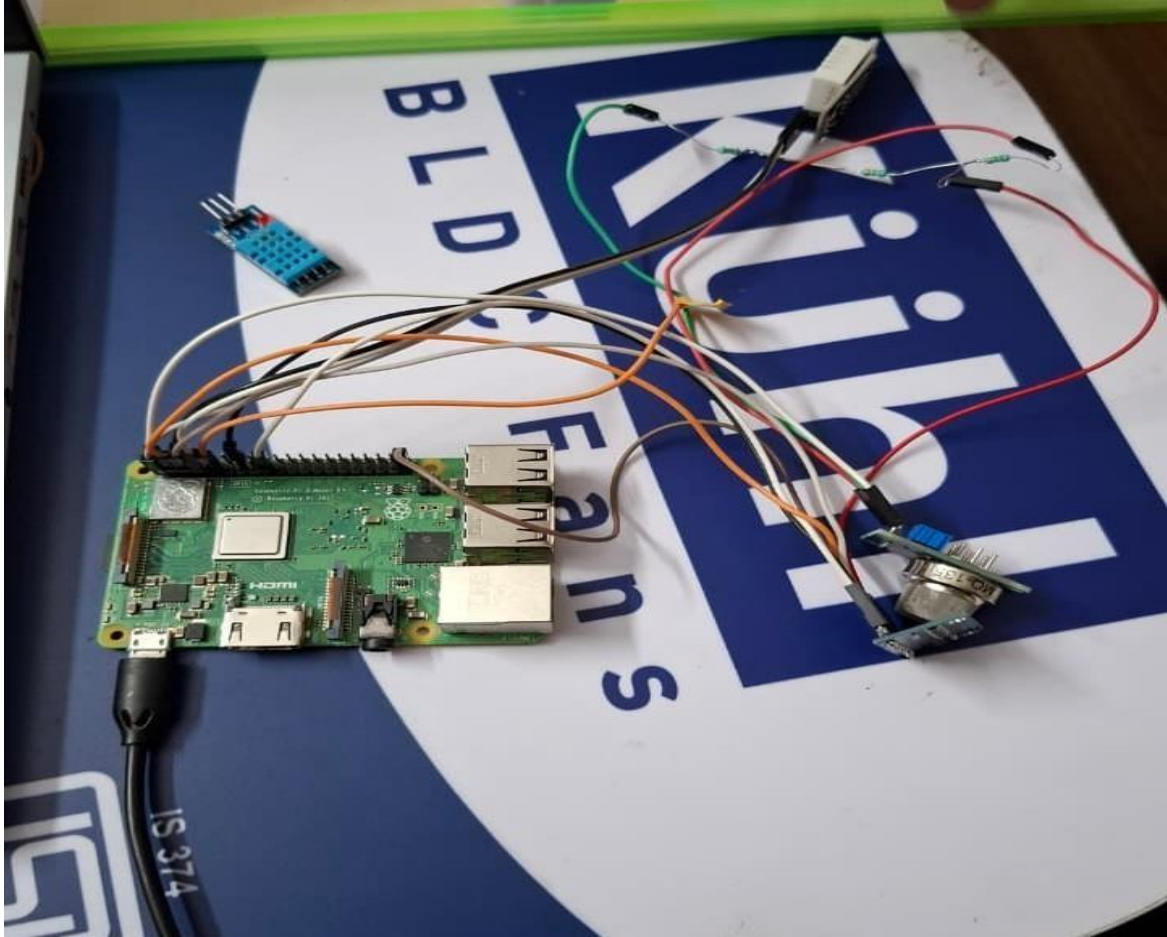


Fig:- Hardware

Conclusion

The Smart Environmental Monitoring System using IoT and Data Analytics provides an efficient solution for real-time monitoring of environmental parameters. The system successfully integrates sensors with a microcontroller to measure temperature, humidity, air quality, and light intensity. The collected data is processed and sent to a cloud platform, enabling remote access and easy visualization.

This project overcomes the limitations of traditional manual monitoring by offering continuous, real-time data access from any location. The implementation of threshold-based alerts improves safety by notifying users when environmental conditions exceed safe limits.

The system shows stable performance in data collection and transmission, making it suitable for applications such as smart homes, industries, agriculture, and pollution monitoring. However, limitations like moderate sensor accuracy and dependence on internet connectivity still exist.

Overall, the project demonstrates the practical use of IoT in environmental monitoring and provides a strong foundation for future smart and automated systems.

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THANK YOU